

ECON 351 · Focused Practice Set

Set 4 of 10

Chapter 4

Elasticity / Types of Goods

3 questions in this practice set

Questions: Q7, Q8, Q9

Screenshots from original HW, sample questions, and mock midterm.

Question 1 of 3 (Practice Q7)

Elasticity #1 · HW1 Ex 4

Exercise 4 - Price Elasticity

Here, we will review the concept of elasticity and go through some examples. Price elasticity of demand is how much quantity demanded changes when the price goes up by 1%. Reminder $E_p = \frac{P}{Q} \frac{\partial Q}{\partial P}$

Take the following demand function $q = 100 - 5p$.

4.1 Calculate the quantity demanded when $p = 5$ and when $p = 10$. What happens to the quantity demanded when the price increases? Given your answer, do you expect the sign of price elasticity to be positive or negative?

4.2 Calculate the price elasticity when $p = 5$. Interpret the elasticity.

4.3 Calculate the quantity demanded when $p = 5.05$.

4.4 Note that $p = 5.05$ corresponds to a 1% price increase with respect to $p = 5$. Compare this % increase with the % change in the quantity demanded. That is, calculate $\Delta Q/Q(p = 5) * 100$. How does this compare to the price elasticity you calculated before?

4.5 Redo points 4.2-4.4 but now calculate the elasticity at $p = 10$ and compare the % change in quantity demanded when $p = 10.1$. How does the % change in quantity compared to the elasticity at $p = 10$?

4.6 EXTRA In the previous questions, you have calculated the elasticity using the formula and also by "brute force". In these cases, the values coincide, but in other cases, there would be a small gap between both methods. Why is that the case, and why are they the same here? (Hard question. Tip: it has to do with the basic concept of a derivative and the curvature of the demand function.)

Question 2 of 3 (Practice Q8)

Elasticity #2 · Sample Q39

Question 38:

A consulting company researched the market for apples in California, and concluded the following: apples and bananas are complements, and apples are an inferior good.

Define the following:

Q_A : quantity of apples demanded

P_A : price of Apples

P_B : price of bananas

I : income

From the options below, mark the only demand function that can possibly represent the demand for apples in California.

(A) $Q_A = 100 - 3P_A + 3I - 5P_B$

(B) $Q_A = 200 - P_A + 2I + 4P_B$

(C) $Q_A = 110 - 2P_A - 4I + 2P_B$

(D) $Q_A = 250 - 4P_A - 5I - 3P_B$

(E) $Q_A = 150 + P_A + I + P_B$

Question 39:

The demand for apples (Q_A) is given by the function

$$Q_A = 784 - 2P_A - 3I - 4P_O$$

where P_A is the price of apples, I is the income level, and P_O is the price of oranges.

Suppose $P_A = 2$, $I = 100$, and $P_O = 20$.

a) Compute the price elasticity of demand for apples.

b) Compute the income elasticity of demand for apples. Are apples a normal good or inferior good?

c) Compute the cross-price elasticity of demand for apples. Are apples and oranges substitutes or complements?

Question 40:

The income elasticity of demand for good X is -2. Due to a financial crisis, the income of consumers declined by 2%. How did the financial crisis affect the quantity demanded for X?

(A) The quantity demanded increased 2%

(B) The quantity demanded decreased 2%

(C) The quantity demanded increased 4%

(D) The quantity demanded decreased 4%

(E) The quantity demanded did not change

Question 41:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 50%, that is, $r = 0.5$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ? Is this consumer saving or borrowing? How much?

Question 42:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 5%, that is, $r = 0.05$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

Question 3 of 3 (Practice Q9)

Elasticity #3 · Sample Q38

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